
Ultra-Wideband Indoor Positioning System

Design Document: Motivation, Objectives, Methodology & Mathematics

Target Platform: **Makerfabs ESP32 UWB Pro (DW1000)**

Sensing: **UWB Ranging + BNO085 IMU**

Output: **XYZ Position + Roll/Pitch/Yaw Orientation**

Version	2.0
Date	June 2026
Status	Active Development

Contents

1	Introduction and Motivation	2
1.1	Problem Statement	2
1.2	Ultra-Wideband as a Candidate Technology	2
1.3	Limitations of Existing Open-Source Libraries	2
1.4	Proposed Contribution	3
2	System Overview	3
2.1	Hardware	3
2.2	Software Architecture	3
3	Objectives	4
4	Mathematical Foundations	4
4.1	Double-Sided Two-Way Ranging (DS-TWR)	4
4.1.1	Motivation	4
4.1.2	Message Exchange	4
4.1.3	ToF Formula	5
4.1.4	Antenna Delay Calibration	5
4.2	NLOS Detection via Channel Impulse Response	5
4.2.1	Physical Basis	5
4.2.2	DW1000 Diagnostic Registers	6
4.2.3	NLOS Score Computation	6
4.2.4	Range Bias under NLOS	6
4.3	Weighted Multilateration	6
4.3.1	Measurement Model	6
4.3.2	Weighted Least-Squares Formulation	7
4.3.3	Robust Outlier Gating	7
4.3.4	Covariance Approximation	7
4.3.5	GPS-Style Simultaneous Bias Estimation	7
4.4	State Estimation	8
4.4.1	Extended Kalman Filter	8
4.4.2	Moving Horizon Estimator	9
4.5	IMU Fusion and Orientation Estimation	9
4.5.1	Quaternion Representation	10
4.5.2	Quaternion Kinematics	10
4.5.3	IMU-Driven Position Propagation	10
4.5.4	Loose vs Tight Coupling	10
4.5.5	Roll, Pitch, and Yaw Extraction	10
4.6	Anchor Self-Survey via Multidimensional Scaling	11
4.6.1	Problem Statement	11
4.6.2	Classical Multidimensional Scaling	11
4.6.3	Coordinate Frame Fixation	11
4.6.4	Noise Propagation	11
5	Implementation Roadmap	11
6	Expected Performance	12

1 Introduction and Motivation

1.1 Problem Statement

Accurate indoor positioning is a prerequisite for autonomous robotics, asset tracking, human-motion capture, augmented reality, and industrial safety systems. The Global Navigation Satellite System (GNSS) fails indoors because satellite signals are attenuated by building materials and the reflected multipath environment renders pseudorange measurements unreliable. Optical systems (cameras, LiDAR) achieve high accuracy but require significant infrastructure, computational resources, and are sensitive to lighting conditions and occlusion. Inertial systems drift unboundedly without external correction.

The result is a gap: no low-cost, infrastructure-light technology currently provides real-time 3D position *and* 3D orientation (six degrees of freedom, 6-DOF) indoors at centimetre-level accuracy with off-the-shelf microcontroller hardware.

1.2 Ultra-Wideband as a Candidate Technology

Ultra-wideband (UWB) radio occupies a spectrum of at least 500 MHz bandwidth (typically 3.1–10.6 GHz for IEEE 802.15.4a devices). The key physical properties that make UWB suitable for ranging are:

- **Time resolution.** The large bandwidth produces a narrow autocorrelation peak. The Decawave DW1000 timestamps signal arrivals at 15.65 ps resolution (64 GHz system clock), corresponding to a theoretical distance resolution of $15.65 \times 10^{-12} \times 3 \times 10^8 \approx 4.7$ mm per clock tick.
- **Multipath separation.** The short pulse duration (~ 2 ns) allows the receiver to distinguish the direct-path component from reflections that arrive within a few nanoseconds. Narrowband systems (Bluetooth, Wi-Fi, Zigbee) cannot make this separation.
- **Material penetration.** UWB signals propagate through common building materials (drywall, wood, glass), enabling ranging through soft obstructions at the cost of a positive distance bias.
- **Low power.** The FCC Part 15 power limit (-41.3 dBm/MHz) combined with the wide bandwidth results in pulse energy well below noise floor for narrowband receivers, making UWB inherently covert and interference-resistant.

The Decawave DW1000 chip, used in the Makerfabs ESP32 UWB Pro module, implements IEEE 802.15.4-2011 UWB and achieves a specified ranging accuracy of ± 10 cm in line-of-sight (LOS) conditions.

1.3 Limitations of Existing Open-Source Libraries

Five open-source Arduino libraries implement DW1000 positioning. All share structural limitations that prevent scaling beyond four anchors or a single mobile tag:

The four-anchor ceiling in `thotro` and its derivatives arises from the RANGE frame encoding: each device occupies 17 bytes (2-byte address + three 5-byte absolute timestamps) within a 90-byte frame buffer. With the 9-byte MAC header, only $\lfloor (90 - 9 - 2) / 17 \rfloor = 4$ devices fit before buffer overflow. The `pizzo00` fork reduces this to 12 bytes/device using differential timestamps, reaching 6 anchors, but the broadcast POLL mechanism still causes RF collision between anchor replies.

Critically, *no existing library* uses the DW1000's first-path power registers for non-line-of-sight (NLOS) detection, despite the chip making this data available on every received frame.

Table 1: Open-source DW1000 library comparison

Feature	thotro	Makerfabs	jremington	pizzo00	dw1000-ng
Anchor ceiling	4	4	7*	6	Unlimited
RF collisions	Yes	Yes	Yes	Yes	No
Update rate (Hz)	0.6	0.6	0.6	1	2
Multi-tag	No	No	No	Partial	No
NLOS detection	No	No	No	No	No
3D positioning	No	No	Fragile	No	No
IMU fusion	No	No	No	No	No
Maintained	No	Partial	Yes	Yes	No

1.4 Proposed Contribution

This project designs and implements a scalable UWB real-time locating system (RTLS) with the following distinguishing characteristics:

1. **Unlimited anchor scaling** via individual addressed DS-TWR exchanges (no shared frame buffer, no broadcast collisions).
2. **NLOS detection** using the DW1000 first-path power registers (FP_AMPL1/2/3) — the first open-source implementation of this capability.
3. **Physics-aware state estimation** using a Moving Horizon Estimator with Huber loss, physical room-boundary constraints, and speed limits.
4. **Full 6-DOF output** (XYZ position + RPY orientation) via tight fusion of UWB ranging and BNO085 IMU data.
5. **Anchor self-survey** via anchor-to-anchor ranging and Multidimensional Scaling, eliminating the need for manual coordinate measurement.

2 System Overview

2.1 Hardware

Hardware Specification

UWB module	Makerfabs ESP32 UWB Pro with Display
UWB chip	Decawave DW1000 (IEEE 802.15.4-2011 UWB)
MCU	ESP32 (dual-core Xtensa LX6, 240 MHz, Wi-Fi/BT)
RF front-end	PA/LNA amplifier (~200 m outdoor range)
Display	1.3 inch SSD1306 OLED (I ² C, pins SDA=4, SCL=5)
IMU (planned)	Bosch BNO085 via SPI
Configuration	3–4 anchors (fixed) + 1 tag (mobile)
Transport	Wi-Fi UDP to MATLAB host (192.168.x.x:5005)

2.2 Software Architecture

The system is split into two layers:

Firmware (Arduino/ESP32): The tag performs round-robin DS-TWR ranging to each anchor, collects raw distances with quality metrics, and streams measurement packets over UDP. Anchors are purely passive responders. The tag never solves its own position.

Host (MATLAB): All position computation, filtering, visualisation, and logging runs on the host PC. This separation means adding anchors requires only editing a JSON configuration file — no firmware change. The firmware packet format is versioned and IMU-ready.

The processing pipeline on the host is:

Raw ranges $\xrightarrow{\text{median filter}}$ Filtered ranges $\xrightarrow{\text{NLOS weighting}}$ Weighted ranges $\xrightarrow{\text{LM multilat.}}$ Position $\xrightarrow{\text{EKF/MHE}}$ $\hat{\mathbf{x}}$ $\xrightarrow{\text{EMA}}$

3 Objectives

- O1.** Achieve XY positioning accuracy of ≤ 3 cm ($1\text{-}\sigma$) in LOS and ≤ 8 cm in soft NLOS conditions using 3–4 anchors.
- O2.** Achieve Z-axis positioning accuracy of ≤ 5 cm with 4 anchors deployed at different heights.
- O3.** Deliver continuous RPY orientation output at ≥ 50 Hz from BNO085 IMU fusion.
- O4.** Maintain position tracking continuity during brief anchor occlusions (≤ 3 s) using IMU dead-reckoning.
- O5.** Support anchor self-survey: determine anchor positions to ≤ 2 cm without external measurement.
- O6.** Demonstrate update rate ≥ 3 Hz for 3 anchors and ≥ 8 Hz in fast-mode operation.
- O7.** Provide real-time NLOS detection per anchor using DW1000 channel impulse response diagnostics.
- O8.** Remain compatible with low-cost hardware: total cost $< \$200$ for a 4-anchor + 1-tag system.

4 Mathematical Foundations

4.1 Double-Sided Two-Way Ranging (DS-TWR)

4.1.1 Motivation

Simple one-way ranging computes distance from the time of flight (ToF) of a single transmitted pulse. This requires synchronised clocks between transmitter and receiver. Crystal oscillators in consumer devices have frequency offsets of ± 20 ppm; at 7 m range, this alone introduces a ranging error of ± 0.7 mm per millisecond of response latency — growing to centimetres with longer delays.

Double-Sided Two-Way Ranging eliminates the clock-offset term analytically through a four-message exchange that gives the receiver two independent estimates of the one-way ToF.

4.1.2 Message Exchange

The DS-TWR protocol involves four timestamped events across tag (T) and anchor (A):

Step	Direction	Frame	Timestamps captured
1	T $\rightarrow$ A	POLL	t_1 (T tx), t_2 (A rx)
2	A $\rightarrow$ T	POLL-ACK	t_3 (A tx), t_4 (T rx)
3	T $\rightarrow$ A	RANGE	t_5 (T tx); payload: t_1, t_4, t_5
4	A $\rightarrow$ T	RANGE-REPORT	t_6 (A rx); anchor computes & replies d

4.1.3 ToF Formula

Define the following intervals (all in DW1000 clock ticks, 15.65 ps each):

$$R_a = t_4 - t_1 \quad (\text{tag: total round-trip, poll to ack-received}) \quad (1)$$

$$D_a = t_3 - t_2 \quad (\text{anchor: processing delay, poll-rx to ack-tx}) \quad (2)$$

$$R_b = t_6 - t_3 \quad (\text{anchor: total round-trip, ack to range-received}) \quad (3)$$

$$D_b = t_5 - t_4 \quad (\text{tag: processing delay, ack-rx to range-tx}) \quad (4)$$

The asymmetric DS-TWR formula (Decawave APS013) gives the time-of-flight:

$$\tau_{\text{ToF}} = \frac{R_a R_b - D_a D_b}{R_a + R_b + D_a + D_b} \quad (5)$$

Clock-offset cancellation. Let $\varepsilon_T, \varepsilon_A$ be the fractional frequency offsets of the tag and anchor crystals respectively. Including these offsets, the true one-way ToF τ satisfies:

$$R_a = (2\tau + D_a)(1 + \varepsilon_T), \quad R_b = (2\tau + D_b)(1 + \varepsilon_A)$$

Substituting into eq. (5) and expanding to first order in ε , the frequency-offset terms cancel exactly, leaving:

$$\tau_{\text{ToF}} \approx \tau + O(\varepsilon^2)$$

Residual error scales as $\varepsilon^2 \approx (20 \text{ ppm})^2 = 4 \times 10^{-10}$, contributing $< 0.1 \text{ mm}$ of ranging error for exchange times $\leq 10 \text{ ms}$.

The measured distance is:

$$\hat{d} = c \cdot \tau_{\text{ToF}} + b_{\text{ant}} \quad (6)$$

where $c = 2.998 \times 10^8 \text{ m/s}$ and b_{ant} is the per-board antenna delay calibration offset.

4.1.4 Antenna Delay Calibration

The DW1000 applies a programmable antenna delay to the RX timestamp register. If the true distance is d^* and the raw measured distance is $\hat{d}(\Delta)$ for delay value Δ , calibration solves:

$$\Delta^* = \arg \min_{\Delta} \left(\mathbb{E} [\hat{d}(\Delta)] - d^* \right)^2$$

via binary search. The relationship is monotone: increasing Δ shifts the RX timestamp later, reducing the computed ToF and thus the reported distance. A binary search over $\Delta \in [15800, 16900]$ with 200 averaged measurements per step converges to the optimal delay within ± 3 ticks ($\approx \pm 1.4 \text{ mm}$) in 14 iterations.

4.2 NLOS Detection via Channel Impulse Response

4.2.1 Physical Basis

The DW1000 correlates the received signal against a known preamble sequence to produce a Channel Impulse Response (CIR) estimate $h(\tau)$. In LOS conditions the CIR has a single dominant peak at the direct-path delay τ_0 . In NLOS conditions (blocked or obstructed direct path), reflected components at delays $\tau_0 + \delta_k$ contribute additional energy to the CIR:

$$h(\tau) = \alpha_0 \delta(\tau - \tau_0) + \sum_{k=1}^K \alpha_k \delta(\tau - \tau_0 - \delta_k)$$

The total received power integrates over all components; the first-path power reflects only the direct component.

4.2.2 DW1000 Diagnostic Registers

After each receive, the DW1000 populates:

- **FP_AMPL1, FP_AMPL2, FP_AMPL3** (register 0x12, offsets 0x07–0x0C): amplitude of the three samples around the first-path leading edge.
- **RXPACC** (register 0x10, offset 0x2C): accumulated path count (preamble accumulator length).
- **CIR_PWR** (register 0x12, offset 0x00): total channel power from the accumulator.

4.2.3 NLOS Score Computation

The first-path power (Decawave APS006 formula) is:

$$P_{\text{FP}} = 10 \log_{10} \left(\frac{A_1^2 + A_2^2 + A_3^2}{N_{\text{acc}}^2} \right) - A_{\text{const}} \quad (7)$$

where A_1, A_2, A_3 are the FP_AMPL register values, N_{acc} is RXPACC, and $A_{\text{const}} = 121.74$ dBm for 16 MHz PRF or 113.77 dBm for 64 MHz PRF.

The total received power is computed from CIR_PWR by the existing driver function `DW1000.getReceivePower()`.

The NLOS score is the power differential:

$$\Delta P_{\text{NLOS}} = P_{\text{RX}} - P_{\text{FP}} \quad (8)$$

Interpretation:

- $\Delta P_{\text{NLOS}} < 3$ dB: LOS. Direct path dominates. Full measurement weight.
- $3 \leq \Delta P_{\text{NLOS}} < 6$ dB: Soft NLOS. Partial obstruction or light multipath. Reduce weight.
- $\Delta P_{\text{NLOS}} \geq 6$ dB: Hard NLOS. Reflected path dominant. Strong downweighting or rejection.

4.2.4 Range Bias under NLOS

NLOS ranging error is always a *positive* bias: reflected and through-wall paths are geometrically longer than the direct path. Through-wall propagation in typical building materials also slows the signal by a factor $\sqrt{\varepsilon_r}$ where ε_r is the relative permittivity (concrete: $\varepsilon_r \approx 6$ –9; drywall: $\varepsilon_r \approx 2$ –4). The empirical range bias b_{NLOS} correlates with ΔP_{NLOS} :

$$b_{\text{NLOS}} \approx k_{\text{NLOS}} \cdot \max(0, \Delta P_{\text{NLOS}} - 3 \text{ dB}) \quad (9)$$

with $k_{\text{NLOS}} \approx 0.02$ – 0.05 m/dB (environment-dependent, tunable). This correction is applied before multilateration.

4.3 Weighted Multilateration

4.3.1 Measurement Model

Given N anchors at known positions $\mathbf{a}_i \in \mathbb{R}^d$ ($d = 2$ or 3) and measured ranges $\hat{d}_i$ (after median filtering and bias correction), the tag position $\mathbf{x} \in \mathbb{R}^d$ satisfies:

$$\hat{d}_i = \|\mathbf{x} - \mathbf{a}_i\| + e_i, \quad e_i \sim \mathcal{N}(0, \sigma_i^2) \quad (10)$$

The noise standard deviation σ_i is range-quality-dependent:

$$\sigma_i = \sigma_{\text{base}} \cdot 10^{\Delta P_{\text{NLOS},i}/20} \quad (11)$$

so that a 10 dB NLOS score increases the assumed noise by a factor of $\sqrt{10} \approx 3.16$ and reduces the weight by a factor of 10.

4.3.2 Weighted Least-Squares Formulation

The weighted multilateration objective is:

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x} \in \mathbb{R}^d} \sum_{i=1}^N w_i \left(\|\mathbf{x} - \mathbf{a}_i\| - \hat{d}_i \right)^2 \quad (12)$$

with weights $w_i = 1/\sigma_i^2$.

This is a nonlinear least-squares problem solved by Levenberg-Marquardt (LM). Define the residual vector $\mathbf{r}(\mathbf{x}) \in \mathbb{R}^N$ with $r_i = \|\mathbf{x} - \mathbf{a}_i\| - \hat{d}_i$, the Jacobian $\mathbf{J} \in \mathbb{R}^{N \times d}$ with rows:

$$\mathbf{J}_i = \frac{\partial r_i}{\partial \mathbf{x}} = \frac{(\mathbf{x} - \mathbf{a}_i)^\top}{\|\mathbf{x} - \mathbf{a}_i\|} \quad (13)$$

and the weight matrix $\mathbf{W} = \text{diag}(w_1, \dots, w_N)$.

The LM update step is:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \left(\mathbf{J}^\top \mathbf{W} \mathbf{J} + \lambda \text{diag}(\mathbf{J}^\top \mathbf{W} \mathbf{J}) \right)^{-1} \mathbf{J}^\top \mathbf{W} \mathbf{r}(\mathbf{x}_k) \quad (14)$$

The damping parameter λ is increased when a step increases cost and decreased otherwise, interpolating between gradient descent (large λ) and Gauss-Newton (small λ).

4.3.3 Robust Outlier Gating

After a first LM solve, outlier anchors are identified and gated. The residuals $\{r_i\}$ are assessed against a scaled median absolute deviation (MAD):

$$\hat{\sigma}_{\text{robust}} = 1.4826 \cdot \text{median}(|r_i - \text{median}(r_i)|) \quad (15)$$

Anchors with $|r_i - \text{median}(r_i)| > K \cdot \hat{\sigma}_{\text{robust}}$ (default $K = 2.5$) are excluded and the solve is repeated with the remaining anchors. The factor 1.4826 makes the MAD a consistent estimator of σ under a Gaussian assumption.

4.3.4 Covariance Approximation

The position covariance is approximated from the Hessian of the gated objective at the solution:

$$\mathbf{P}_{\text{pos}} \approx \sigma_{\text{base}}^2 \left(\mathbf{J}^\top \mathbf{W} \mathbf{J} \right)^{-1} \quad (16)$$

This is passed to the Kalman filter as the measurement covariance $\mathbf{R}_k$.

4.3.5 GPS-Style Simultaneous Bias Estimation

When $N \geq d + 2$ anchors are available, the augmented state $\tilde{\mathbf{x}} = [\mathbf{x}^\top, \delta]^\top \in \mathbb{R}^{d+1}$ is solved, where δ is a common range bias (absorbing tag delay miscalibration and average multipath):

$$\hat{d}_i = \|\mathbf{x} - \mathbf{a}_i\| + \delta + e_i \quad (17)$$

The augmented Jacobian appends a column of ones:

$$\tilde{\mathbf{J}}_i = [\mathbf{J}_i, 1] \quad (18)$$

This is directly analogous to GPS pseudorange processing, where the receiver clock offset appears as a common additive term on all satellite measurements.

Geometric note. Solving for δ with $N = d + 1$ anchors (minimum for standard multilateration) is equivalent to Time Difference of Arrival (TDOA) positioning, since δ cancels in all pairwise range differences. TDOA is geometrically weaker than TOA for symmetric anchor layouts — the problem becomes ill-conditioned near the anchor centroid. This mode is therefore only enabled when $N \geq d + 2$, providing at least one degree of overdetermination.

4.4 State Estimation

4.4.1 Extended Kalman Filter

The EKF models the tag as a constant-acceleration (CA) particle. The state vector is:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \\ \mathbf{a} \\ \delta \end{bmatrix} \in \mathbb{R}^{3d+1} \quad (19)$$

where $\mathbf{p} \in \mathbb{R}^d$ is position, $\mathbf{v} \in \mathbb{R}^d$ is velocity, $\mathbf{a} \in \mathbb{R}^d$ is acceleration, and $\delta \in \mathbb{R}$ is the slowly-varying range bias.

The discrete-time transition model with time step Δt is:

$$\mathbf{x}_{k+1} = \mathbf{F}\mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}) \quad (20)$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_d & \Delta t \mathbf{I}_d & \frac{\Delta t^2}{2} \mathbf{I}_d & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_d & \Delta t \mathbf{I}_d & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_d & \mathbf{0} \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (21)$$

The process noise covariance for the CA model is (Singer model):

$$\mathbf{Q} = q_j \begin{bmatrix} \frac{\Delta t^5}{20} \mathbf{I} & \frac{\Delta t^4}{8} \mathbf{I} & \frac{\Delta t^3}{6} \mathbf{I} & \mathbf{0} \\ \frac{\Delta t^4}{8} \mathbf{I} & \frac{\Delta t^3}{3} \mathbf{I} & \frac{\Delta t^2}{2} \mathbf{I} & \mathbf{0} \\ \frac{\Delta t^3}{6} \mathbf{I} & \frac{\Delta t^2}{2} \mathbf{I} & \Delta t \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & q_\delta \Delta t \end{bmatrix} \quad (22)$$

where q_j is the jerk power spectral density (tuning parameter) and q_δ governs the rate of bias drift.

The measurement model for anchor i is (tight coupling):

$$h_i(\mathbf{x}) = \|\mathbf{p} - \mathbf{a}_i\| + \delta \quad (23)$$

with Jacobian row:

$$\mathbf{H}_i = \left[\frac{(\mathbf{p} - \mathbf{a}_i)^\top}{\|\mathbf{p} - \mathbf{a}_i\|}, \mathbf{0}_{1 \times 2d}, 1 \right] \quad (24)$$

The standard Kalman prediction and update equations are:

$$\text{Predict: } \hat{\mathbf{x}}_{k|k-1} = \mathbf{F}\hat{\mathbf{x}}_{k-1|k-1} \quad (25)$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\mathbf{P}_{k-1|k-1}\mathbf{F}^\top + \mathbf{Q} \quad (26)$$

$$\text{Update: } \mathbf{S}_k = \mathbf{H}_k\mathbf{P}_{k|k-1}\mathbf{H}_k^\top + \mathbf{R}_k \quad (27)$$

$$\mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}_k^\top\mathbf{S}_k^{-1} \quad (28)$$

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \left(\mathbf{z}_k - h(\hat{\mathbf{x}}_{k|k-1}) \right) \quad (29)$$

$$\mathbf{P}_{k|k} = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_{k|k-1} \quad (30)$$

4.4.2 Moving Horizon Estimator

The MHE solves a constrained nonlinear optimisation over a sliding window of N past measurement epochs. At time k , the window spans $\{k - N + 1, \dots, k\}$.

Objective function:

$$\min_{\{\mathbf{x}_t\}_{t=k-N+1}^k} \underbrace{\|\mathbf{x}_{k-N+1} - \bar{\mathbf{x}}_{k-N+1}\|_{\mathbf{P}_{\text{arr}}^{-1}}^2}_{\text{arrival cost}} + \underbrace{\sum_{t=k-N+2}^k \|\mathbf{x}_t - \mathbf{F}\mathbf{x}_{t-1}\|_{\mathbf{Q}^{-1}}^2}_{\text{dynamics residuals}} + \underbrace{\sum_{t=k-N+1}^k \sum_{i=1}^{N_a} \rho_\delta(\|\mathbf{p}_t - \mathbf{a}_i\| - \hat{d}_{t,i})}_{\text{range residuals}} \quad (31)$$

where $\|\mathbf{v}\|_{\mathbf{M}}^2 = \mathbf{v}^\top \mathbf{M} \mathbf{v}$, $\bar{\mathbf{x}}_{k-N+1}$ and $\mathbf{P}_{\text{arr}}$ are the prior mean and covariance from the previous window solution (arrival cost captures past information compactly), and ρ_δ is the Huber loss function:

$$\rho_\delta(e) = \begin{cases} \frac{1}{2}e^2 & \text{if } |e| \leq \delta \\ \delta\left(|e| - \frac{\delta}{2}\right) & \text{if } |e| > \delta \end{cases} \quad (32)$$

with threshold $\delta = 0.15$ m (one standard deviation of typical ranging noise). Large residuals from NLOS anchors grow linearly rather than quadratically, reducing their influence without complete rejection.

Constraints:

$$\mathbf{x}_{\min} \leq \mathbf{p}_t \leq \mathbf{x}_{\max} \quad (\text{room boundary, hard}) \quad (33)$$

$$\|\mathbf{v}_t\|_2 \leq v_{\max} \quad (\text{speed limit, hard}) \quad (34)$$

The boundary constraint prevents the estimator from producing physically impossible positions during anchor occlusion. The speed limit (e.g. $v_{\max} = 2$ m/s for walking) eliminates large transient jumps.

Computational cost. With $N = 6$, $d = 3$, and $N_a = 4$ anchors, the optimisation has $N \times (3d + 1) = 60$ decision variables and is solved by `fmincon` (interior-point method) in approximately 2–5 ms per step, well within the ~ 333 ms UWB sweep period.

Comparison with EKF:

Table 2: EKF vs MHE comparison for this application

Property	EKF	MHE
NLOS handling	Pre-filtering required	Huber loss absorbs outliers
Physical constraints	None	Room boundary, speed limit (hard)
Nonlinearity	Linearised via Jacobian	Solved exactly
Lag	Zero (one sample)	$N/2$ samples (≈ 1 s at $N = 6$)
Computation per step	$O(d^2)$, trivial	2–5 ms (fmincon)
Sensitivity to init	Moderate	Low (window smooths errors)

4.5 IMU Fusion and Orientation Estimation

4.5.1 Quaternion Representation

Orientation is represented by a unit quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]^\top \in \mathbb{R}^4$ satisfying $\|\mathbf{q}\| = 1$. The rotation matrix from body frame to world frame is:

$$\mathbf{R}(\mathbf{q}) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix} \quad (35)$$

4.5.2 Quaternion Kinematics

The continuous-time quaternion differential equation driven by body angular rate $\boldsymbol{\omega}^b = [\omega_x, \omega_y, \omega_z]^\top$ is:

$$\dot{\mathbf{q}} = \frac{1}{2} \mathbf{q} \otimes \begin{bmatrix} 0 \\ \boldsymbol{\omega}^b \end{bmatrix} \quad (36)$$

where $\otimes$ denotes the Hamilton product. The discrete propagation over Δt is:

$$\mathbf{q}_{k+1} = \mathbf{q}_k \otimes \exp\left(\frac{\Delta t}{2} \begin{bmatrix} 0 \\ \boldsymbol{\omega}_k^b \end{bmatrix}\right) \approx \mathbf{q}_k \otimes \begin{bmatrix} \cos(\frac{\Delta t}{2} \|\boldsymbol{\omega}_k^b\|) \\ \frac{\boldsymbol{\omega}_k^b}{\|\boldsymbol{\omega}_k^b\|} \sin(\frac{\Delta t}{2} \|\boldsymbol{\omega}_k^b\|) \end{bmatrix} \quad (37)$$

4.5.3 IMU-Driven Position Propagation

The BNO085 outputs linear acceleration $\mathbf{a}^b$ (gravity removed) in the body frame. The world-frame acceleration is $\mathbf{a}^w = \mathbf{R}(\mathbf{q})\mathbf{a}^b$. Position and velocity propagate as:

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t + \frac{1}{2} \mathbf{a}_k^w \Delta t^2 \quad (38)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}_k^w \Delta t \quad (39)$$

This propagation runs at the IMU rate (100 Hz), providing 33 intermediate position estimates between each UWB measurement update (3 Hz). UWB range measurements correct the accumulated drift at each available sweep.

4.5.4 Loose vs Tight Coupling

Loose coupling treats the multilaterated position as the measurement. The EKF state is updated with $\mathbf{z}_k = \hat{\mathbf{p}}_{\text{multilat}}$ and $\mathbf{R}_k = \mathbf{P}_{\text{pos}}$ from eq. (16). This is simpler to implement but suffers from the two-stage sequential estimation error.

Tight coupling uses the raw UWB ranges directly as measurements via the nonlinear model of eq. (23). This avoids the intermediate multilateration step, correctly propagates uncertainty, and allows the filter to use partial anchor sets (e.g. 2 of 4 anchors visible) without changing the measurement model structure.

4.5.5 Roll, Pitch, and Yaw Extraction

The Euler angles (roll ϕ , pitch θ , yaw ψ) in ZYX convention are extracted from the quaternion as:

$$\phi = \text{atan2}\left(2(q_w q_x + q_y q_z), 1 - 2(q_x^2 + q_y^2)\right) \quad (40)$$

$$\theta = \arcsin(2(q_w q_y - q_z q_x)) \quad (41)$$

$$\psi = \text{atan2}\left(2(q_w q_z + q_x q_y), 1 - 2(q_y^2 + q_z^2)\right) \quad (42)$$

The BNO085 ARVR Stabilised Rotation Vector mode applies sensor fusion (accelerometer + gyroscope + magnetometer) internally at 400 Hz and outputs a quaternion robust to magnetic disturbances. The magnetometer provides an absolute yaw reference, preventing the unbounded yaw drift that affects gyroscope-only integration.

4.6 Anchor Self-Survey via Multidimensional Scaling

4.6.1 Problem Statement

Given N anchors, measure all $\binom{N}{2}$ pairwise distances $\hat{d}_{ij}$ by having each anchor temporarily act as a ranging initiator. Recover the N anchor positions $\{\mathbf{a}_i\}$ that are consistent with these distances.

4.6.2 Classical Multidimensional Scaling

Construct the squared-distance matrix $\mathbf{D}^{(2)} \in \mathbb{R}^{N \times N}$ with $D_{ij}^{(2)} = \hat{d}_{ij}^2$. Apply double-centering via the centering matrix $\mathbf{J} = \mathbf{I}_N - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$:

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J} \quad (43)$$

If the true distances were exact, $\mathbf{B}$ would be a rank- d positive semidefinite matrix satisfying $\mathbf{B} = \mathbf{X}\mathbf{X}^\top$ where $\mathbf{X} \in \mathbb{R}^{N \times d}$ contains the true centred positions. With noisy distances, $\mathbf{B}$ is full rank. The best rank- d approximation is obtained from the eigendecomposition $\mathbf{B} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^\top$ by taking the d largest eigenvalues:

$$\hat{\mathbf{X}} = \mathbf{V}_d\mathbf{\Lambda}_d^{1/2} \quad (44)$$

where $\mathbf{V}_d \in \mathbb{R}^{N \times d}$ and $\mathbf{\Lambda}_d = \text{diag}(\lambda_1, \dots, \lambda_d)$ contain the d dominant eigenpairs.

4.6.3 Coordinate Frame Fixation

The MDS solution is defined only up to a rigid-body transformation (rotation, translation, reflection). Fix the coordinate frame by convention:

1. **Translation:** Subtract $\hat{\mathbf{x}}_1$ (row 1 of $\hat{\mathbf{X}}$) from all rows so anchor 1 is at the origin.
2. **Rotation:** Rotate so that $\hat{\mathbf{x}}_2 - \hat{\mathbf{x}}_1$ aligns with the positive X axis.
3. **Reflection:** If anchor 3 lies in the negative Y half-plane, negate all Y coordinates.

The resulting layout is written to `config/anchors.json` and used for all subsequent localisation without modification.

4.6.4 Noise Propagation

Each pairwise distance is the mean of $M = 100$ ranging measurements. If individual measurement noise is σ_d , the mean has uncertainty $\sigma_d/\sqrt{M}$. For $\sigma_d \approx 5$ cm and $M = 100$, the per-pair uncertainty is 0.5 cm. Classical MDS propagates this to anchor position uncertainty of approximately $\sigma_d/\sqrt{M}$ in each coordinate — consistent with the objective of ≤ 2 cm anchor self-survey accuracy.

5 Implementation Roadmap

The system is developed in six phases, each independently testable:

Table 3: Implementation phases and key items

Phase	Item	Description	Difficulty
1 — Foundation	1.0	Switch to 64 MHz PRF (accuracy mode)	Easy
	1.1	Better calibration (200 samples, multi-distance)	Easy
	1.2	Adaptive polling: skip dead anchors	Easy
	1.3	Faster sweep rate (reduce reply delay)	Medium
	1.5	Anchor self-survey + auto <code>anchors.json</code>	Medium
2 — Ranging	2.1	FP_POWER NLOS detection (firmware + wire format)	Medium
	2.2	Weighted multilateration (NLOS score)	Medium
	2.3	NLOS range bias correction slider	Easy
	2.4	Per-anchor diagnostics panel	Easy
	2.5	GPS-style δ -solve (4+ anchors)	Medium
3 — Estimation	3.1	Constant-acceleration EKF	Medium
	3.2	δ as EKF state (bias estimation)	Med-Hard
	3.3	Moving Horizon Estimator	Hard
	3.4	Hybrid EKF + MHE	Medium
4 — 3D + RPY	4.1	4th anchor + 3D configuration	Easy
	4.2	BNO085 SensorImu (SPI)	Medium
	4.3	RPY output from quaternion	Easy
	4.4	Loose coupling: IMU-aided EKF	Hard
	4.5	Tight coupling: raw ranges + IMU in MHE	Very Hard
5 — Hardening	5.1	Deep sleep on anchors	Medium
	5.2	DW1000 driver error handling	Medium
	5.3	Temperature compensation	Medium
	5.4	Online self-calibration	Hard
6 — Scale	6.1	Multi-tag TDMA superframe	Hard
	6.2	Differential timestamp encoding	Hard
	6.3	On-device 2D solve (OLED)	Medium

Table 4: Expected accuracy progression through implementation phases

After Phase	XY accuracy	Z accuracy	RPY	Rate	Key enabler
Current	3–10 cm	—	—	3 Hz	Baseline
Phase 1	2–6 cm	—	—	4–8 Hz	Calibration + PRF
Phase 2	1–4 cm	—	—	4–8 Hz	NLOS detection
Phase 3	1–3 cm	—	—	4–8 Hz	MHE + CA-EKF
Phase 4	1–3 cm	2–5 cm	3–5°	100 Hz RPY	IMU fusion
Phase 5	1–3 cm	2–5 cm	3–5°	100 Hz RPY	Reliability

6 Expected Performance

The theoretical floor for the DW1000 chip is approximately 1–2 cm in perfect LOS conditions (limited by 15.65 ps timestamp resolution and residual crystal asymmetry). Practical indoor environments with multipath and occasional NLOS will see 2–5 cm as the sustainable floor. IMU integration does not improve the UWB ranging accuracy but eliminates position lag between sweeps and provides orientation at sensor rate.